

CS661 Project — Dataset Reference Document

GH Archive Dataset Summary

1 File Information

Property	Value
File name	<code>gh_archive_data.csv</code>
File format	CSV (comma-separated values)
File size	485.3 MB
Total rows	3,448,426
Total columns	10
Date range	2023-01-01 00:03:48 to 2024-12-31 23:58:44
Observation window	2 full years (Jan 2023 – Dec 2024)

2 Data Source

The data was extracted from **GH Archive** (<https://gharchive.org>), which records every public GitHub event since 2011 in structured JSON format with hourly granularity. We accessed this dataset through the **Google BigQuery public dataset** (`githubarchive.*`) and queried only the event types and repositories relevant to our project. The extracted data was downloaded as a CSV file for local processing.

3 Column Descriptions

Column	Data Type	Description
<code>repo</code>	string	Full repository name in the format <code>owner/repo-name</code> , e.g., <code>facebook/react</code> .
<code>date</code>	datetime	Timestamp of when the event occurred on GitHub. Format: <code>YYYY-MM-DD HH:MM:SS (UTC)</code> .
<code>event_type</code>	string	The type of GitHub event. One of six possible values: <code>PullRequestEvent</code> , <code>IssuesEvent</code> , <code>IssueCommentEvent</code> , <code>PushEvent</code> , <code>WatchEvent</code> , <code>ReleaseEvent</code> .
<code>actor</code>	string	GitHub username of the person or bot who triggered the event. Bot accounts typically contain keywords like <code>bot</code> , <code>[bot]</code> , or <code>robot</code> in their username.

<code>action</code>	string	The specific action within an event type. For example, a <code>PullRequestEvent</code> can have action <code>opened</code> , <code>closed</code> , or <code>reopened</code> . For <code>PushEvent</code> and <code>WatchEvent</code> , this field contains <code>none</code> and <code>started</code> respectively.
<code>pr_or_issue_number</code>	integer	The numeric identifier of the pull request or issue associated with the event. For <code>PushEvent</code> and <code>WatchEvent</code> , this is 0 or irrelevant.
<code>closed_at</code>	datetime	Timestamp of when the issue or pull request was closed. If not closed, this is stored as <code>1970-01-01 00:00:00</code> , which is a placeholder for null and must be treated as missing during preprocessing.
<code>merged_at</code>	datetime	Timestamp of when a pull request was merged. If the PR was not merged (closed without merge), this is stored as <code>1970-01-01 00:00:00</code> (placeholder for null). Only meaningful for <code>PullRequestEvent</code> rows.
<code>state</code>	string	The current state of the issue or pull request at the time of the event. One of: <code>open</code> , <code>closed</code> , or <code>none</code> (for events like <code>PushEvent</code> that have no state).
<code>author_association</code>	string	The relationship of the actor to the repository at the time of the event. One of: <code>OWNER</code> , <code>MEMBER</code> , <code>COLLABORATOR</code> , <code>CONTRIBUTOR</code> , <code>NONE</code> .

4 Repositories Covered

The dataset covers **30 repositories** across three technology ecosystems.

4.1 Frontend Ecosystem — 9 repositories, 651,205 events

Repository	Description
<code>facebook/react</code>	React JavaScript UI library
<code>facebook/react-native</code>	React framework for mobile apps
<code>vuejs/vue</code>	Vue.js progressive JS framework
<code>sveltejs/svelte</code>	Svelte compiler-based JS framework
<code>angular/angular</code>	Angular TypeScript framework
<code>vercel/next.js</code>	Next.js React production framework
<code>tailwindlabs/tailwindcss</code>	Tailwind CSS utility framework
<code>mui/material-ui</code>	Material UI component library
<code>twbs/bootstrap</code>	Bootstrap CSS framework

4.2 ML/Data Ecosystem — 9 repositories, 1,345,003 events

Repository	Description
pytorch/pytorch	PyTorch deep learning framework
tensorflow/tensorflow	TensorFlow ML framework
keras-team/keras	Keras deep learning library
huggingface/transformers	HuggingFace Transformers library
microsoft/DeepSpeed	DeepSpeed distributed training
ray-project/ray	Ray distributed computing
scikit-learn/scikit-learn	Scikit-learn ML library
pandas-dev/pandas	Pandas data analysis library
numpy/numpy	NumPy numerical computing

4.3 Backend/DevOps Ecosystem — 11 repositories, 1,160,021 events

Repository	Description
microsoft/vscode	Visual Studio Code editor
kubernetes/kubernetes	Kubernetes container orchestration
golang/go	Go programming language
elastic/elasticsearch	Elasticsearch search engine
apache/spark	Apache Spark data processing
hashicorp/terraform	Terraform infrastructure as code
ansible/ansible	Ansible automation platform
tiangolo/fastapi	FastAPI Python web framework
django/django	Django Python web framework
pallets/flask	Flask Python web framework
docker/compose	Docker Compose tool

5 Event Type Distribution

Event Type	Row Count	Percentage
IssueCommentEvent	1,555,472	45.1%
PushEvent	582,564	16.9%
WatchEvent	573,744	16.6%
PullRequestEvent	411,496	11.9%
IssuesEvent	323,307	9.4%
ReleaseEvent	1,843	0.1%
Total	3,448,426	100%

6 Action Field Values by Event Type

The action field means different things depending on the `event_type`:

Event Type	Possible Action Values
PullRequestEvent	opened, closed, reopened
IssuesEvent	opened, closed, reopened
IssueCommentEvent	created
WatchEvent	started
PushEvent	none
ReleaseEvent	published

Overall action distribution across all rows:

Action	Count
created	1,555,471
none	582,565
started	573,744
opened	361,092
closed	356,319
reopened	17,392
published	1,843

7 Events Per Repository

Repository	Event Count
pytorch/pytorch	686,671
microsoft/vscode	321,380
kubernetes/kubernetes	300,200
flutter/flutter	292,197
tensorflow/tensorflow	177,584
vercel/next.js	167,952
huggingface/transformers	163,464
golang/go	153,747
elastic/elasticsearch	149,284
facebook/react-native	125,694
ray-project/ray	101,050
angular/angular	78,263
facebook/react	74,958
mui/material-ui	66,999
apache/spark	66,954
pandas-dev/pandas	63,473
sveltejs/svelte	61,385
microsoft/DeepSpeed	47,092
scikit-learn/scikit-learn	41,658
numpy/numpy	39,186
hashicorp/terraform	36,837
tiangolo/fastapi	36,758
tailwindlabs/tailwindcss	34,890
ansible/ansible	33,663
django/django	30,532
twbs/bootstrap	25,230
keras-team/keras	24,825
docker/compose	19,549
vuejs/vue	15,834
pallets/flask	11,117

8 Actor and Contributor Statistics

Property	Value
Total unique actors	520,041
Unique human actors	519,233
Unique bot actors	808

8.1 Author Association Distribution

The `author_association` field indicates the contributor's relationship to the repository:

Association	Count	Meaning
NONE	2,081,429	External user, no formal association
CONTRIBUTOR	868,940	Has had a PR merged previously
MEMBER	376,365	Member of the owning organization
COLLABORATOR	120,911	Explicitly granted write access
OWNER	781	Owner of the repository

9 Bot Activity

A significant portion of events are generated by automated bot accounts. Bots are identified by the presence of keywords `bot`, `[bot]`, `robot`, or `mirror` in the actor username.

Property	Value
Total bot events	848,909
Total human events	2,599,517
Bot percentage	24.62%
Unique bot accounts	808

9.1 Top 15 Bot Accounts by Event Count

Bot Username	Event Count
k8s-ci-robot	148,901
pytorchmergebot	126,308
github-actions[bot]	105,592
copybara-service[bot]	104,577
facebook-github-bot	70,241
pytorch-bot[bot]	61,508
gopherbot	58,464
VSCoDeTriageBot	32,464
auto-submit[bot]	29,505
fluttermirroringbot	9,986
angular-automatic-lock-bot[bot]	9,480
k8s-triage-robot	9,207
vs-code-engineering[bot]	8,529
google-ml-butler[bot]	5,782
mui-bot	5,383

10 Pull Request Statistics

Property	Value
Total PullRequestEvent rows	411,496
Unique PR numbers across all repos	140,024
Matched open-to-merge PR pairs	98,518
Median time to merge	11.5 hours
Mean time to merge	149.8 hours
Minimum time to merge	<1 hour
Maximum time to merge	16,020.8 hours ($\approx$ 668 days)

Note on computing PR latency: PR latency must be computed by matching a `PullRequestEvent` with `action=opened` to a corresponding `PullRequestEvent` with

`action=closed` and a non-null `merged_at`, joined on `repo` and `pr_or_issue_number`. The `date` field of the opened event gives the opened timestamp; `merged_at` gives the merge timestamp. The large gap between median (11.5 hrs) and mean (149.8 hrs) indicates the presence of long-lived PRs that skew the average.

11 Issue Statistics

Property	Value
Total IssuesEvent rows	323,307
Issues opened	157,887
Issues closed	152,908
Issues reopened	12,512
Unique issue numbers across all repos	148,576
Issues with at least one comment	134,101
Median time to first response	3.9 hours
Mean time to first response	235.5 hours

Note on computing issue response time: Response time is the difference between the `date` of an `IssuesEvent` with `action=opened` and the earliest `date` of an `IssueCommentEvent` on the same `repo` and `pr_or_issue_number`. The high mean (235.5 hrs) vs. median (3.9 hrs) indicates that many issues receive rapid responses while a smaller number sit unattended for weeks or months.

12 Monthly Distribution

Property	Value
Average events per month	143,684
Minimum events in a month	109,807 (December 2023)
Maximum events in a month	166,339 (October 2024)

13 State Field Distribution

State	Count
open	1,657,457
closed	632,817
none	1,158,152

`none` corresponds to event types that do not have a state concept (`PushEvent`, `WatchEvent`, `ReleaseEvent`).

14 Known Data Issues and How to Handle Them

14.1 1970-01-01 Placeholder Dates

The `merged_at` and `closed_at` fields use 1970-01-01 00:00:00 as a placeholder when the value is absent (i.e., a PR was not merged, or an issue was not closed). These must be converted to `NaN/None` during preprocessing before any time-based computation.

```
df.loc[df['merged_at'].dt.year == 1970, 'merged_at'] = None
df.loc[df['closed_at'].dt.year == 1970, 'closed_at'] = None
```

14.2 PR Latency Cannot Be Read Directly

The `merged_at` timestamp in a `PullRequestEvent` row records when the PR was merged but does not tell you when it was opened. To compute latency you must join the opened event and the closed event for the same PR number within the same repo.

14.3 Bot Inflation

24.62% of all events are bot-generated. Any metric computed over raw event counts (commit frequency, PR volume, comment rate) will be inflated unless bots are filtered. Always filter using:

```
df['is_bot'] = df['actor'].str.contains(
    'bot|Bot|\\[bot\\]|robot|mirror', regex=True
)
```

14.4 IssueCommentEvent Covers Both Issues and PRs

GitHub fires `IssueCommentEvent` for comments on both issues and pull requests. If you want to analyse only issue comments, filter by joining against `IssuesEvent` numbers. If you want only PR comments, join against `PullRequestEvent` numbers.

14.5 flutter/flutter Not in Our Ecosystem List

The repo `flutter/flutter` appears in the data with 292,197 events but was not explicitly listed in our proposal ecosystems. It falls under the Frontend ecosystem loosely. The team should decide whether to include it in frontend analysis or exclude it. It is in the data either way.

15 Mapping to Visualization Tasks

Visualization		Columns / Fields Used
Technology Adoption Streamgraph		<code>repo</code> , <code>date</code> , <code>event_type</code> — aggregate monthly event counts per repo/ecosystem.
Contributor Network Graph		<code>actor</code> , <code>repo</code> , <code>pr_or_issue_number</code> , <code>event_type=IssueCommentEvent</code> — connect actors who commented on the same PR. Filter <code>is_bot=False</code> .
PR Lifecycle Sankey + Box Plots		<code>event_type=PullRequestEvent</code> , <code>action</code> , <code>date</code> , <code>merged_at</code> , <code>closed_at</code> , <code>pr_or_issue_number</code> , <code>repo</code> .
Issue Responsiveness Heatmap		<code>event_type=IssuesEvent</code> and <code>event_type=IssueCommentEvent</code> , <code>date</code> , <code>repo</code> , <code>pr_or_issue_number</code> .
Bot vs. Human Bar Chart		<code>actor</code> , <code>is_bot</code> (derived), <code>repo</code> , <code>event_type</code> , <code>date</code> .
Repository Health Dashboard		All derived metrics: median merge time, bus-factor score, issue response rate, bot percentage — computed from the above.

16 Preprocessing Checklist

Before building any visualization, the following preprocessing steps must be completed:

1. Load `gh_archive_data.csv` into a pandas `DataFrame`.

2. Parse `date`, `merged_at`, `closed_at` as datetime objects.
3. Replace 1970-01-01 values in `merged_at` and `closed_at` with `None`.
4. Add `is_bot` boolean column using actor name pattern matching.
5. Add `ecosystem` column mapping each repo to Frontend, ML/Data, or Backend/DevOps.
6. Add `year_month` column derived from `date` for monthly aggregations.
7. Save the cleaned DataFrame to a local SQLite database (`github_analytics.db`) for fast querying during visualization development.